DGMS 2019 • SECTION L: HYDROGEOLOGY & ENVIRONMENTAL GEOLOGY

Question 88: Coastal Aquifer Hydrogeology: Ghyben-Herzberg Saltwater Intrusion

Official Examination Guide • Freshwater Lens Dynamics, Upconing & Anthropogenic Encroachment

OFFICIAL EXAMINATION QUESTION

In coastal areas, groundwater resource is being threatened by the encroachment of saltwater because:

1. Freshwater being less dense than saltwater, floats over the latter and forms large lens-shaped body

2. Saltwater being dense, is available at very shallow depths in coastal regions

3. Heavy pumping in coastal areas causes encroachment of saltwater

4. Waves and tides cause mixing of saltwater with freshwater

Which of the statements given above are correct?

(a) 1 and 3 only CORRECT ANSWER

(b) 2 and 4 only

(c) 1, 3 and 4 only

(d) 1, 2, 3 and 4

1. Executive Summary & Core Hydrogeological Principles

In coastal groundwater systems (Todd & Mays, *Groundwater Hydrology*; Fetter, *Applied Hydrogeology*), the delicate equilibrium between terrestrial freshwater and adjacent marine saltwater is governed by the **Ghyben-Herzberg Principle (1888, 1901)**:

- **Freshwater Lens Floatation (Statement 1 is TRUE):** Pure terrestrial freshwater has a lower density ($\rho_f \approx 1.000\text{ g/cm}^3$) than mineralized seawater ($\rho_s \approx 1.025\text{ g/cm}^3$). Consequently, recharge water naturally floats above the denser saltwater wedge, forming a convex unconfined freshwater lens. Under static equilibrium:

  ```
  z = [ ρf / (ρs - ρf) ] · h ≈ [ 1.000 / (1.025 - 1.000) ] · h = 40 · h
  ```

  where h is the elevation of the freshwater table above mean sea level (MSL), and z is the depth of the freshwater-saltwater interface below sea level.

- **Heavy Pumping & Upconing (Statement 3 is TRUE):** When heavy extraction occurs in coastal wellfields, local drawdown lowers the water table (Δh). Because of the 40:1 hydrostatic balance ratio, **a 1-meter drawdown in the water table causes the underlying saltwater interface to rise by approximately 40 meters**! This rapid upward migration (known as *saltwater upconing*) and inland lateral advance directly salinizes coastal production wells.

- **Statement 2 is FALSE:** In healthy coastal aquifers with natural inland recharge, the freshwater lens maintains a deep protective wedge, depressing the saltwater interface tens to hundreds of meters below sea level ($z = 40h$); saltwater is NOT situated at "very shallow depths" inland.

- **Statement 4 is NOT the Driving Mechanism:** Although oceanic tides create localized mechanical dispersion along the immediate shoreline beach zone, this does not threaten regional groundwater resources—the actual threat is regional upconing and lateral encroachment driven by anthropogenic over-abstraction.

Thus, Statements **1 and 3 only** are the valid explanatory reasons, making Option **(a)** the correct choice.

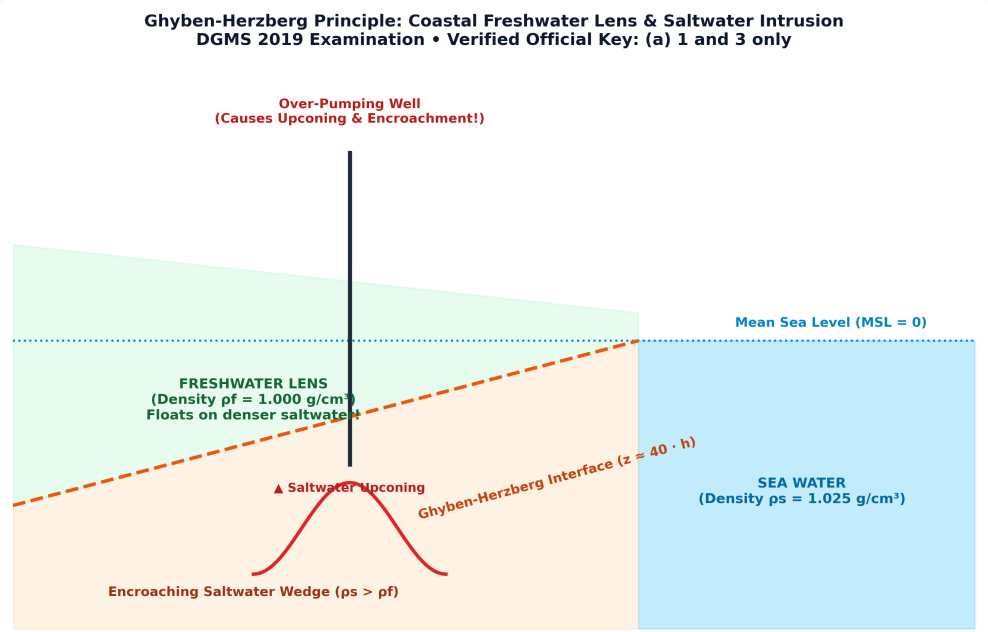


Figure 88.1: Cross-sectional hydrogeological model of a coastal aquifer illustrating the Ghyben-Herzberg freshwater lens ($z \approx 40h$) and the catastrophic 40:1 saltwater upconing induced by heavy well pumping (Grounded in Todd & Mays).

2. Mathematical Derivation of the Ghyben-Herzberg Relation

Consider a column of groundwater extending from the water table down to the interface point at depth z below sea level:

```
Pressure of freshwater column at interface = Pressure of
seawater column at interface
Pf = Ps
ρf · g · (h + z) = ρs · g · z
ρf · h + ρf · z = ρs · z
z · (ρs − ρf) = ρf · h
z = [ ρf / (ρs − ρf) ] · h

For typical seawater (ρs = 1.025 g/cm³) and freshwater
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(ρf = 1.000 g/cm³):
z = [ 1.000 / 0.025 ] · h = 40 · h
```

3. Evaluation of Options

- **Statement 1: CORRECT.** Density difference ($\rho_f < \rho_s$) causes freshwater to float atop saltwater, forming a lens.
- **Statement 2: INCORRECT.** Saltwater occurs at depth ($40h$ below sea level), not at very shallow depths unless the freshwater lens has already been completely destroyed.
- **Statement 3: CORRECT.** Excessive pumping causes 40:1 upconing, driving severe saltwater encroachment into coastal wells.
- **Statement 4: INCORRECT AS REGIONAL THREAT.** Tidal action only affects the immediate surficial beach boundary; it does not drive regional basin-scale encroachment.

> **Key Takeaway:** Ghyben-Herzberg relation ($z \approx 40h$): 1 m drop in water table causes 40 m rise of saltwater. Density floatation (1) combined with overpumping (3) threatens coastal aquifers. Hence, **(a) 1 and 3 only** is the correct answer.

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